

Co-Producing Climate Resilience: A Regional Hub for Equitable Solutions to Septic System Failures

Theme and Rationale. In many rural and coastal regions of the Mid-Atlantic and Southeastern United States, wastewater infrastructure is quietly, and dangerously, failing beneath the surface. Aging and climate-stressed septic systems are reaching a breaking point, with rising groundwater tables, intensified flooding, and increasingly saturated soils triggering widespread system failures. The impacts are cascading: nitrate-contaminated drinking water, nutrient-overloaded estuaries, health risks for frontline residents, and deepening infrastructure inequities, among others. These failures are especially acute in historically underserved communities that are more reliant on private wells and septic systems. In many of these communities, planning tools are outdated, financing mechanisms are fragmented, and the people most affected are too often excluded from decision-making.

This project proposes not only to respond to this crisis but to reimagine how communities plan, finance, and adapt critical water infrastructure under climate stress. We aim to establish a foundation for a **Regional Hub for Climate Resilience and Equitable Wastewater Solution**, a transdisciplinary, community-anchored platform that brings together researchers, practitioners, policymakers, and residents to co-create scalable models for adaptive wastewater transitions. Rather than addressing septic system failure as an isolated technical issue, we frame it as a systems challenge that cuts across public health, climate equity, environmental governance, and intergenerational infrastructure planning. At the heart of this effort is a commitment to shared ownership of both problems and solutions. The hub will initially be anchored in Virginia's Eastern Shore, where trusted relationships and community priorities have already been established through our ongoing work. From this starting point, the project aims to strengthen relationships and collaborative capacity between researchers and communities in coastal Virginia and Maryland, building the foundation for a broader regional expansion. Our long-term vision is to scale this model across other low-lying and climate-vulnerable regions, including Delaware and the Carolinas, enabling a regionally coordinated approach to climate-resilient wastewater planning that is rooted in equity and grounded in place.

By treating the Eastern Shore as a living laboratory for co-producing knowledge and action, we will prototype adaptive, community-centered solutions that can guide transitions from failing systems to climate-ready infrastructure across the broader Mid-Atlantic and Southeastern coastal region. In our proposed research, community engagement is not an add-on; it is the foundation that will guide refining the research, interpreting results, and shaping solutions. Our methodology explicitly resists extractive models of research and instead centers co-creation at every stage. Our research will be organized into four integrated research thrusts:

- **Thrust 1: Real-Time Vulnerability Mapping through AI and Community-Integrated Sensing.** We will develop a real-time, AI-enabled framework to identify and monitor septic system vulnerabilities in the Eastern Shore of Virginia. This framework will integrate low-cost sensors (e.g., groundwater levels, system saturation), community-sourced reporting tools, and high-resolution environmental modeling and data to detect early warning signs of failure. The resulting dynamic vulnerability maps will merge technical and experiential knowledge, helping communities visualize risk patterns under current and projected climate conditions.
- **Thrust 2: Decision Support for Adaptive Wastewater Transitions.** This thrust will co-design a user-friendly decision-support platform to help local governments, utilities, and residents evaluate decentralized wastewater alternatives. Drawing on the outputs of Thrust 1, the platform will visualize tradeoffs among clustered systems, enhanced onsite technologies, and nature-based options, considering factors such as soil suitability,

groundwater risk, household costs, and community preferences. We will pilot early-stage versions of this platform with stakeholders in the Eastern Shore and identify opportunities for cross-site replication.

- **Thrust 3: Laying the Groundwork for Understanding Cascading Environmental and Public Health Risks.** This thrust will initiate foundational work to assess how failing septic systems contribute to nutrient contamination in groundwater and surface waters, particularly in low-lying coastal areas where shallow aquifers intersect with septic drainfields. We will compile and digitize existing septic and well records, identify key data gaps, and collect limited pilot data in areas with suspected contamination. These efforts will inform the design of more comprehensive sampling, exposure modeling, and risk communication strategies in future phases, supporting both environmental health research and regulatory dialogue.
- **Thrust 4: Laying the Groundwork for Equitable Financing and Scalable Implementation.** We will begin exploring policy, institutional, and market barriers to implementing decentralized solutions at scale. This includes reviewing current funding and permitting pathways, mapping equity challenges related to ownership and affordability, and identifying opportunities for public-private collaboration. Engagement with the Environmental Finance Center Network and other regional stakeholders will help define pathways for future development of household incentives, shared infrastructure financing models, and scalable investment strategies.

Current Work This Builds Upon. This project builds directly on our NSF Coastlines and People Climate Equity (CoPe) project, which has cultivated deep partnerships and trust in the Eastern Shore of Virginia. Through this work, residents have consistently identified failing septic systems, contaminated wells, and infrastructure inaccessibility as urgent and overlooked climate resilience challenges. This proposed initiative is a direct response to that call. It operationalizes community-driven resilience planning by advancing implementable solutions, not just diagnostics. In addition to deepening this work, we will formally collaborate with Dr. Allison Riley at the University of Maryland, who led an EPA-funded study on climate-induced septic system failure, allowing us to align methodologies, share community engagement strategies, and build on their existing infrastructure and partnerships. Their network of nonprofit, agency, and community partners across Maryland will enrich cross-site learning and expand the hub's reach. We will also collaborate with Dr. Molly Mitchell at the Virginia Institute of Marine Science, whose applied research on septic suitability, public health, and sea level rise impacts in coastal communities brings critical expertise in environmental health risk, geospatial analysis, and stakeholder engagement. Her experience bridging technical research with planning practice and community priorities will directly inform our work in the Eastern Shore and strengthen cross-institutional coordination in coastal Virginia.

Project Team. The project is led by Majid Shafiee-Jood (Engineering), with co-leadership from Barbara Brown Wilson (Architecture/the Virginia Equity Center) and Teresa Culver (Engineering). Together, this core team brings expertise in climate adaptation, infrastructure planning, groundwater modeling, environmental equity, and community-engaged research. Our project team also includes Allison Riley (Univ. Maryland) and Molly Mitchell (VIMS). By engaging both Dr. Mitchell and Dr. Riley as expert partners, we will lay the groundwork for expanding this work across additional regions in coastal Virginia and Maryland, building a strong foundation for a truly regional approach. In parallel, we will begin forging new partnerships in the Carolinas and Delaware to identify shared priorities, data needs, and pathways for collaboration. Through this project, therefore, we aim to lay the groundwork for a nationally replicable model of climate-resilient infrastructure transformation that is guided by science and powered by

collaboration to establish UVA as a leader in coordinated, community-anchored responses to one of the most overlooked climate adaptation challenges facing rural America.

Anticipated Budget Expenses. We anticipate a total budget of \$500,000 over two years. The budget will support faculty summer salary, undergraduate and graduate research assistants, and postdoctoral researchers to lead and implement the research thrusts. A portion of the funds will be allocated to the acquisition and deployment of sensor equipment for monitoring septic system performance and water quality, including support for retrofitting existing systems in collaboration with participating households. We will also dedicate funding to support sustained community engagement, including co-design activities, local convenings, and participant incentives that ensure inclusive and equitable involvement throughout the research process.